

Q1: a) Define Overfitting and Underfitting in Supervised learning. For each, give one observable symptom on a learning curve.

- Overfitting: model too complex and it memorize the training data. It perform well on training data but fail when meet unseen test data.

- Underfitting: model too simple to capture the structure of data. It performs poorly both training and validation sets.

Observable symptom:

- Overfitting occurs in the right region: train error decreasing to zero, while validation error start increasing again.

- Underfitting occurs in the left region: both curves are high and close together.

b) A model achieve Train Accuracy = 0,99 and Validation Accuracy = 0,62. Which phenomenon is this?

It is overfitting. The model achieve training accuracy at 0,99 but performs poorly at validation, producing a ~37 percent gap.

Q2: Bias - Variance (Foundation)

Given formular

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

where: $\text{Bias}^2 = (E[\hat{f}(x)] - f(x))^2$ - error from wrong assumptions

$\text{Variance} = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$ - sensitivity to training

sample

σ^2 = irreducible noise in data - generating process

The irreducible term: σ^2 come from randomness in y itself

($y = f(x) + \epsilon$, $E(\epsilon) = 0$, $\text{Var}(\epsilon) = \sigma^2$) and cannot be eliminated by any model.

Q3: Gini and information gain (Decision tree)

a) S has 3B, 5A $\rightarrow$ Compute $G(S)$

Total sample, $|S| = |5 + 3| = 8$

$$P_A = \frac{5}{8} = 0,625; \quad P_B = \frac{3}{8} = 0,375$$

Gini impurity:

$$G(S) = 1 - (P_A^2 + P_B^2) = 1 - (0,625^2 + 0,375^2) = 0,46875$$

b) Split: $S_L(3A, 0B)$; $S_R(2A, 3B)$. Compute weight Gini and gain

Split result

$$S_L: 3A, 0B \rightarrow |S_L| = 3$$

$$S_R: 2A, 3B \rightarrow |S_R| = 5$$

$$\Rightarrow G(S_L) = 1 - \left(\left(\frac{3}{3} \right)^2 + \left(\frac{0}{3} \right)^2 \right) = 0$$

$$G(S_R) = 1 - \left(\left(\frac{2}{5} \right)^2 + \left(\frac{3}{5} \right)^2 \right) = 0,48$$

$$G_{\text{split}} = \frac{|S_L|}{|S|} G(S_L) + \frac{|S_R|}{|S|} G(S_R) = \frac{3}{8} (0) + \frac{5}{8} (0,48) = 0,3$$

Information gain

$$\Delta G = G(S) - G_{\text{split}} = 0,46875 - 0,3 = 0,16875$$

Q4: How CART handles continuous attributes?

1) Sorting: For a continuous feature, its values are sorted in ascending order ($v_1 < v_2 < \dots < v_m$)

2) Midpoints: Candidate thresholds are generated at the midpoints between adjacent distinct values

$$t_i = \frac{v_i + v_{i+1}}{2}$$

3) Best split² Selection: Each candidate threshold partitions the samples into 2 subsets ($x \leq t_i$ and $x > t_i$)

CART computes the impurity reduction for all candidates and selects the threshold that maximizes the gain.